Task 3 Transferring vertical loads

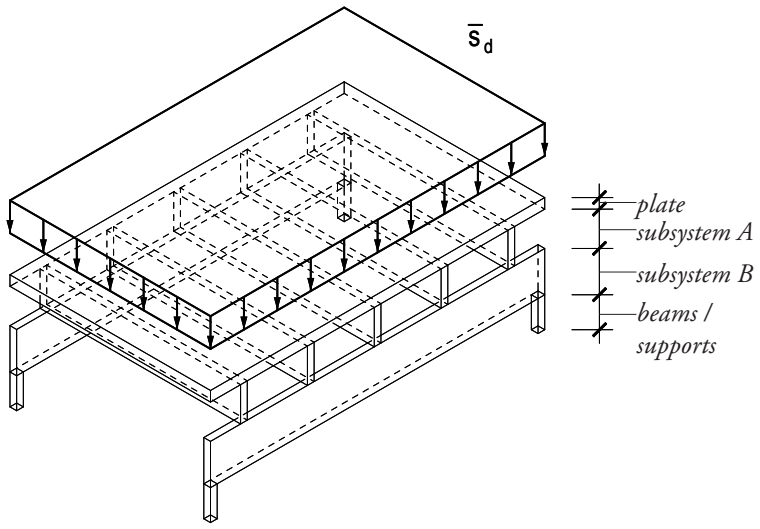
The plate is loaded by a dead area load of $\bar{s}_k$ = 1 kN/m².

a) Calculate the design value of the constant area load. Draw the relevant tributary area for one of the five beams (subsystem A) into the floor plan. Calculate the line load g_d over the relevant beam.

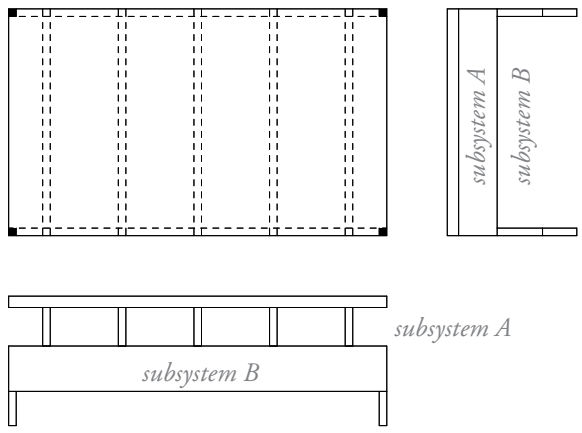
b) Draw a possible internal force flow in the beam (subsystem A) for the line load found in a). Draw the corresponding force diagram. Indicate tension forces with red and compression forces with blue.

c) The reaction forces from b) are the same for all five beams that support the plate. These are further transferred to two longitudinal beams (subsystem B). First draw the applied forces. Secondly, find an internal force flow in one of these longitudinal beams with the aid of the force diagram, considering that the maximum compression force is 310 kN. (Hint: In simple arch-cable structures, the elements at the supports are subjected to the maximum force.) Indicate tension forces with red, compression forces with blue and reaction forces with green.

a)

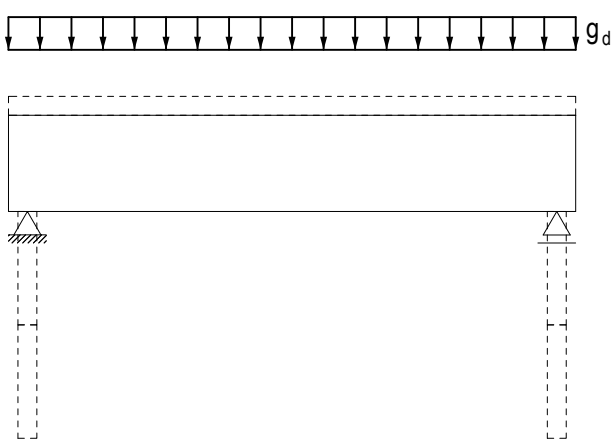


axonometric drawing



plans 1:500

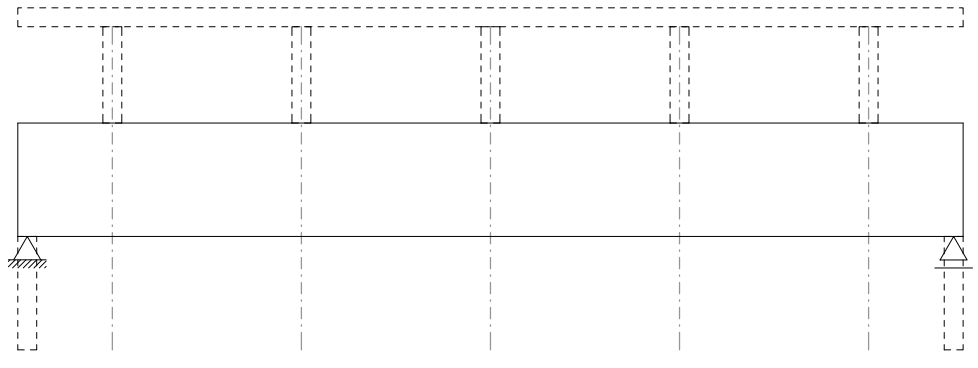
b)



form diagram subsystem A 1:200

force diagram 1cm ≙ 50kN

c)



form diagram subsystem B 1:200

force diagram 1cm ≙ 50kN